

Section : Common

Q.1 If A and B are square matrices of the same order 3, such that $\det(A) = 3$ and $AB = 3I$, where I is an identity matrix of order 3. Then the value of $\det(B)$ is:

1. 3
2. 27
3. 9
4. 1

Options 1. 1
2. 2
3. 3
4. 4

Q.2 The difference of two different skew-symmetric matrices is:

1. Null matrix
2. Identity matrix
3. Symmetric matrix
4. Skew-symmetric matrix

Options 1. 1
2. 2
3. 3
4. 4

Q.3 The general solution of the differential equation $\frac{dy}{dx} = e^{ax+by}$ is: (Here C is an arbitrary constant)

1. $be^{-by} + ae^{ax} = C$
2. $-be^{-by} + ae^{ax} = C$
3. $ae^{-by} + be^{ax} = C$
4. $e^{ax} + e^{by} = C$

Options 1. 1
2. 2
3. 3
4. 4

Q.4 Value of $\int \frac{2}{(x-3)\sqrt{x+1}} dx$ is: (Here C is an arbitrary constant)

1. $\log \left| \frac{x-3}{x+1} \right| + C$
2. $\log \left| \frac{\sqrt{x-1}-2}{\sqrt{x+1}} \right| + C$
3. $\frac{1}{2} \log \left| \frac{\sqrt{x+1}-1}{\sqrt{x+1}+1} \right| + C$

4. $\log \left| \frac{\sqrt{x+1}-2}{\sqrt{x+1}+2} \right| + C$

Options 1. 1

2. 2

3. 3

4. 4

Q.5 The maximum value of $\sin x \cdot \cos x$ is:

1. 1

2. $\frac{1}{2}$

3. $\frac{1}{4}$

4. $\sqrt{2}$

Options 1. 1

2. 2

3. 3

4. 4

Q.6 Consider the Linear Programming Problem

Maximize $z = x + y$

Subject to the constraints $x - y \leq -1, x \geq y, x \geq 0, y \geq 0$

Then which one of the following is TRUE?

1. Maximum $z = 3$ at $(1, 2)$

2. There is no solution

3. Maximum $z = 15$ at $(7, 8)$

4. Maximum $z = 10$ at all points on the line $x - y = -1$

Options 1. 1

2. 2

3. 3

4. 4

Q.7

If $A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & -3 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 3 \\ 4 & -5 \\ 1 & 2 \end{bmatrix}$, then which of the following statements are TRUE?

(A) AB is defined

(B) AB and BA both are defined and $AB = I$, where I is an identity matrix of order 2

(C) BA is defined

(D) AB and BA both are defined and $AB = BA$

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (C) only

2. (B), (C) and (D) only

3. (A) and (C) only

4. (A), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.8

If $e^x + e^y = e^{x+y}$, then $\frac{dy}{dx} =$

1. e^{x-y}

2. e^{y-x}

3. $-e^{y-x}$

4. e^{x+y}

Options 1. 1

2. 2

3. 3

4. 4

Q.9

Match \textbf{List-I} with \textbf{List-II}

\textbf{List-I}

\textbf{List-II}

\textbf{Differential Equation}

\textbf{Order and degree of differential equation}

(A) $\frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^2 = e^{\frac{dy}{dx}} + 1$

(I) Order = 1, Degree = 2

(B) $\left(\frac{d^2y}{dx^2}\right)^2 + 4\left(\frac{dy}{dx}\right)^3 = e^y - 1$

(II) Order = 2, Degree = 1

(C) $3\left(\frac{dy}{dx}\right) + 4y + e^y = \frac{dx}{dy}$

(III) Order = 2, Degree = 2

(D) $\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right) = \left(e^y + \frac{dy}{dx}\right)^2$ (IV) Order = 2, Degree = Not defined

Choose the \textbf{correct} answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

3. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

4. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.10

Function $f(x) = x^3 - 3x + 3$ is

(A) Increasing in the interval $(-1, 1)$

(B) Increasing in the interval $(1, \infty)$

(C) Decreasing in the interval $(-1, 1)$

(D) Increasing in the interval $(-\infty, -1) \cup (1, \infty)$

Choose the \textbf{correct} answer from the options given below:

1. (A), (C) and (D) only

2. (B), (C) and (D) only

3. (A) and (B) only

4. (A), (B) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.11

If $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, then the value of A^{20} is:

1. $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

2. $\begin{bmatrix} 20 & 20 \\ 0 & 20 \end{bmatrix}$

3. $\begin{bmatrix} 1 & 20 \\ 0 & 1 \end{bmatrix}$

4. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Options 1. 1

2. 2

3. 3

4. 4

Q.12

The value of $\int_0^1 xe^x dx$ is:

1. $\frac{1}{e}$

2. e

3. 1

4. -1

Options 1. 1

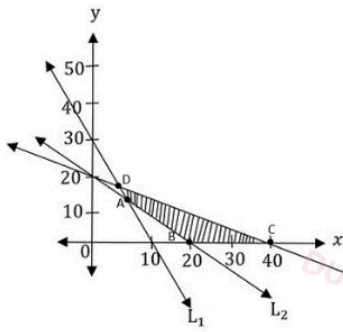
2. 2

3. 3

4. 4

Q.13

Which one of the following inequalities is redundant for the shaded feasible region (ABCD) shown below?



1. $x + 2y \leq 40$
2. $3x + y \geq 30$
3. $4x + 3y \leq 60$
4. $x \geq 0, y \geq 0$

Options 1. 1

2. 2
3. 3
4. 4

Q.14 The area (in sq. units) bounded by the parabola $y^2 = 4ax$, its latus rectum and the x -axis in the first quadrant is:

1. $\frac{1}{3}a^2$
2. $\frac{2}{3}a^2$
3. $\frac{4}{3}a^2$
4. $\frac{8}{3}a^2$

Options 1. 1

2. 2
3. 3
4. 4

Q.15 Let X denotes the number of hours a person uses a mobile and the probability distribution of X is as:

X	0	1	2	3	4
P(X)	0.1	K	2K	2K	K

Then the value of K is

1. 0.15
2. 0.20
3. 0.12
4. 0.16

Options 1. 1

2. 2
3. 3
4. 4

- Q.16** An automobile dealer wishes to buy four luxury cars of different brands with some down payment and balance in equal monthly instalments for 10 years. The bank charges 9% interest per annum compounded monthly.

Given $\frac{0.0075(1.0075)^{120}}{(1.0075)^{120} - 1} = 0.01266$

Luxury Car Price of the Car(in Rs.) Down payment(in Rs.)

P	25,00,000	5,00,000
Q	35,00,000	12,00,000
R	45,00,000	15,00,000
S	42,00,000	15,00,000

Match List I with List II

List-I List-II

Luxury Car EMI(in Rs.)

(A)P	(i) 34,182
(B)Q	(ii) 37,980
(C)R	(iii) 29,118
(D)S	(iv) 25,320

Choose the correct answer from the options given below:

- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
- (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
- (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

- Q.17** The standard deviation of the number of tails in three tosses of a coin is:

- $\frac{3}{4}$
- $\frac{3}{2}$
- $\frac{3}{\sqrt{2}}$
- $\frac{\sqrt{3}}{2}$

Options 1. 1

2. 2

3. 3

4. 4

- Q.18** If a revenue function is given by $R(x) = 2027x - 1013x^2 - 675x^3$, then the marginal revenue function (MR) is:

- $MR = 2027 - 1013x - 675x^2$
- $MR = 2026 - 4050x$
- $MR = 2027 - 2026x - 2025x^2$
- $MR = 1013x - 2025x^2$

Options 1. 1

2. 2

3. 3

4. 4

- Q.19** Match List I with List II

List-I

(Inequality)

List-II

(Solution Set)

(A) $2x - 3 < x + 2 \leq 3x + 5, x \in \mathbb{R}$ (I) $x \in (-1, \infty)$

(B) $|2x + 3| < 7, x \in \mathbb{R}$ (II) $x \in (-\infty, 120]$

(C) $\frac{1}{2} \left(\frac{3}{5}x + 4 \right) \geq \frac{1}{3}(x - 6), x \in \mathbb{R}$ (III) $x \in (-5, 2)$

(D) $\frac{|x+1|}{x+1} > 0, x \in \mathbb{R} - \{-1\}$ (IV) $x \in \left[-\frac{3}{2}, 5\right)$

Choose the correct option

1. (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

2. (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

3. (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

4. (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

Options 1. 1

2. 2

3. 3

4. 4

Q.20 Which of the following are types of statistical inferences?

(A) Point estimation

(B) Interval estimation

(C) Marginal estimation

(D) Hypothesis testing

Choose the correct answer from the options given below:

1. (A), (B) and (D) only

2. (A), (B) and (C) only

3. (A), (B), (C) and (D)

4. (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.21 Three bad eggs are mixed with 7 good ones

If two eggs are drawn one by one without replacement

then the probability distribution of the number X

of bad eggs drawn is

1.)

X 0 1 2

P(X) $\frac{1}{4}$ $\frac{1}{2}$ $\frac{1}{4}$

2.)

X 0 1 2

P(X) $\frac{15}{61}$ $\frac{20}{61}$ $\frac{26}{61}$

3.)

X 0 1 2

P(X) $\frac{7}{15}$ $\frac{7}{15}$ $\frac{1}{15}$

4.)

X 0 1 2

P(X) $\frac{1}{8}$ $\frac{1}{4}$ $\frac{5}{8}$

Options 1. 1

2. 2

3. 3

4. 4

Q.22 The following data shows the percentage of urban Indian households who have a high speed 5G internet connection

Year(x)	2020	2021	2022	2023	2024
Urban housue hold(y)	9%	18%	21%	29%	38%

If a straight line trend by the method of least squares is

$$y = 23 + 6.9(x - 2022)$$

then the forecast for the year 2025 is

1. 54.6%

2. 43.7%

3. 47.6%

4. 34.8%

Options 1. 1

2. 2

3. 3

4. 4

Q.23 Mr. Mittal invested Rs. 20,000 in a mutual fund in the year 2019. The value of the mutual fund increased to Rs. 32,000 in the year 2024. The compound annual growth rate (CAGR) of his investment is:

[Given $(1.6)^{1/5} = 1.098$]

1. 8.5%

2. 10%

3. 9.8%

4. 9.1%

Options 1. 1

2. 2

3. 3

4. 4

Q.24 If 95% confidence interval for the population mean was reported to be 140 to 150 and $\sigma = 25$, then the size of the sample used in this study is:

[Given: $Z_{0.025} = 1.96$]

1. 120

2. 81

3. 96

4. 112

Options 1. 1

2. 2

3. 3

Q.25 The value of a depreciable asset at the end of its useful life is called ____.

1. Nominal Value
2. Book value
3. Marginal value
4. Scrap value

Options 1. 1

2. 2
3. 3
4. 4

Q.26 Which of the following are examples of irregular trends in a time series?

- (A) Decrease in production due to a sudden strike.
- (B) The rise in prices before festivals.
- (C) Unusual rise in income of the printing press due to the announcement of an election.
- (D) Fall in crop yield due to floods.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (C) and (D) only
3. (B), (C) and (D) only
4. (B) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.27 The area of the region bounded by the curves $y = x^2 + 2$ and the x -axis, between $x = 0$ and $x = 3$ in the first quadrant is:

1. $\frac{16}{3}$ sq.units
2. 15 sq.units
3. $\frac{21}{2}$ sq.units
4. 12 sq.units

Options 1. 1

2. 2
3. 3
4. 4

Q.28 Which of the following statements are correct?

- (A) If A is a square matrix, then $|A^2| = |A|^2$.
- (B) If A and B are square matrices of the same order, then $\det(AB) = \det(A) + \det(B)$.
- (C) If A is a square matrix of order 3 and $|A| = 2$, then the value of $|-3A|$ is 54.
- (D) If the matrix $\begin{bmatrix} 5-x & x-1 \\ 3 & 5 \end{bmatrix}$ is singular, then the value of x is $\frac{7}{2}$.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A) and (D) only
4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.29 The unit digit of 12^{12} is:

1. 4

2. 2

3. 0

4. 6

Options 1. 1

2. 2

3. 3

4. 4

Q.30

If $y = \sqrt{x + \sqrt{x + \sqrt{x + \dots}}}$, then

1. $(2y + 1) \frac{dy}{dx} + 3\sqrt{x} = 0$

2. $3 \frac{dy}{dx} + 3y^2 = 0$

3. $3x \frac{dy}{dx} + (2y + 1) = 0$

4. $(2y - 1) \frac{dy}{dx} - 1 = 0$

Options 1. 1

2. 2

3. 3

4. 4

Q.31 A steamer can row at the speed of 16km/hr in still water. If the river is flowing at 8 km/hr and it takes 12 hours for a round trip, then the distance between the two places is:

1. 48 km

2. 72 km

3. 96 km

4. 54 km

Options 1. 1

2. 2

3. 3

4. 4

Q.32 In 5 trials of binomial distribution, the probability of 3 successes is 4 times the probability of 2 successes.

The probability of success in each trial is:

1. $\frac{1}{5}$

2. $\frac{4}{5}$

3. $\frac{3}{4}$

4. $\frac{2}{3}$

Options 1. 1

2. 2

3. 3

4. 4

Q.33 A person has taken a loan of Rs. 40,000 for 3 months from a lender who has deducted Rs. 2,000 as interest at the time of lending. Then the effective rate of interest charged per annum by the lender is:
[Given $(1.0526)^4 = 1.2275$]

1. 10.50%
2. 21%
3. 22.75%
4. 16.75%

Options 1. 1

2. 2
3. 3
4. 4

Q.34 Let $A = \begin{bmatrix} 0 & 2\alpha + 1 \\ 1 & \beta \end{bmatrix}$ and $B = [b_{ij}]$ be a skew symmetric matrix of order 2 such that $b_{12} = 1$.

If $AB = I_2$, where I_2 is the identity matrix of order 2, then:

1. $\alpha + \beta = 1$
2. $\beta - \alpha = 1$
3. $\alpha + \beta = -2$
4. $\alpha\beta = 1$

Options 1. 1

2. 2
3. 3
4. 4

Q.35 Pipe A can fill the tank 3 times faster than pipe B. If both pipes A and B running together can fill the tank in 15 minutes, then the time taken by pipe B alone to fill the tank is:

1. 42 minutes
2. 72 minutes
3. 60 minutes
4. 48 minutes

Options 1. 1

2. 2
3. 3
4. 4

Q.36 A furniture trader deals in only two items — chairs and tables. He has Rs. 50,000 to invest and a space to store at most 35 items. A chair costs Rs. 1000 and a table costs Rs. 2000. The trader earns a profit of Rs. 150 on a chair and Rs. 250 on a table. Choose the correct option that describes the given linear programming problem (LPP) to maximize the profit, where x and y are the number of chairs and tables.

1. Maximize $Z = 150x + 250y$,
Subject to constraints,
 $x + y \leq 35$, $x + 2y \geq 50$, $x \geq 0$, $y \geq 0$
2. Maximize $Z = 150x + 250y$,
Subject to constraints,
 $x + y \leq 35$, $x + 2y \leq 50$, $x \geq 0$, $y \geq 0$
3. Maximize $Z = 150x + 250y$,
Subject to constraints,
 $x + y \geq 35$, $2x + y \leq 50$, $x \geq 0$, $y \geq 0$

4. Maximize $Z = 150x + 250y$,
Subject to constraints,
 $x + y \geq 35$, $2x + y \geq 50$, $x \geq 0$, $y \geq 0$

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.37 If X is a normal variate with mean 16 and standard deviation 4, then the value of the standard normal variate Z corresponding to $X = 17$ is:

- 1. 1.25
- 2. -1.25
- 3. -0.25
- 4. 0.25

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.38 If A is a non-singular matrix of order 3 such that $|\text{adj}(A)| = 121$, then $|AA^T|$ is equal to:

- 1. 121
- 2. 21
- 3. 63
- 4. 11

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.39 If the slope of the tangent to the curve $y = y(x)$ at any point (x, y) is $\frac{2x}{y^2}$ and the curve passes through the point $\left(\frac{1}{\sqrt{3}}, 1\right)$, then the equation of the curve is:

- 1. $y^2 = 3x^3$
- 2. $y^3 = 3x^2$
- 3. $y^4 = 3x^2$
- 4. $y = 3x^2$

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.40 In a 700 m race, the ratio of speeds of two participants A and B is 5 : 6. If A has a start of 150 m, then the distance by which A wins the race is:

- 1. 80 m
- 2. 40 m
- 3. 75 m
- 4. 100 m

Options 1. 1

2. 2

3. 3

4. 4

Q.41 For the given five values 15, 24, 18, 33, 42, the three-year moving averages are:

1. 19, 22, 23

2. 19, 25, 31

3. 19, 30, 31

4. 19, 25, 33

Options 1. 1

2. 2

3. 3

4. 4

Q.42 What sum of money is needed to invest now so as to get Rs. 5000 at the beginning of every month forever, if the money is worth 6% per annum compounded monthly?

1. Rs. 10,05,000

2. Rs. 15,00,000

3. Rs. 10,50,000

4. Rs. 12,50,000

Options 1. 1

2. 2

3. 3

4. 4

Q.43 The value of the definite integral $I = \int_0^2 x\sqrt{2-x} dx$ is:

1. $\frac{16}{15\sqrt{2}}$

2. $\frac{16\sqrt{3}}{15}$

3. $\frac{16\sqrt{2}}{15}$

4. $\frac{5\sqrt{2}}{7}$

Options 1. 1

2. 2

3. 3

4. 4

Q.44 Which of the following is NOT correct about a Sinking Fund?

1. It is set up for particular upcoming expense.

2. A fixed amount at regular intervals is deposited in it.

3. It can be used only for the purpose it was created.

4. It is a long-term fund which can be closed at any time.

Options 1. 1

2. 2

3. 3

4. 4

Q.45

If the matrix $M = \begin{bmatrix} 0 & -1 & 3\alpha \\ 1 & \beta & -5 \\ -6 & 5 & 0 \end{bmatrix}$ is skew-symmetric, then

1. $\alpha = 2, \beta = 1$
2. $\alpha = 2, \beta = -1$
3. $\alpha = 2, \beta = 0$
4. $\alpha = 1, \beta = 0$

Options 1. 1

2. 2
3. 3
4. 4

Q.46

For the function $f(x) = x^{1/x}$, $x > 0$, which of the following are correct?

- (A) $x = 0$ is the only point where extremum may occur.
(B) The given function is maximum at $x = e$.
(C) The function has no extreme value for $x > 0$.
(D) The maximum value of the function $f(x)$ is $e^{1/e}$.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A) and (C) only
3. (C) and (D) only
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.47

In what ratio should a grocery shopkeeper mix two varieties of pulses worth Rs. 85 per kg and Rs. 100 per kg respectively, so as to get a mixture worth Rs. 92 per kg?

1. 8 : 7
2. 5 : 6
3. 3 : 7
4. 11 : 10

Options 1. 1

2. 2
3. 3
4. 4

Q.48

If the system of equations $2x + 3y = 10$ and $x + ky = 4$ has a unique solution, then

1. $k = \frac{3}{2}$
2. $k \neq \frac{3}{2}$
3. $k \neq 0$
4. $k \neq \frac{1}{2}$

Options 1. 1

2. 2
3. 3

Q.49 Consider the following hypothesis test:

$$H_0 : \mu = 18$$

$$H_1 : \mu \neq 18$$

If a sample of 48 provided a sample mean $\bar{x} = 17$ and a sample standard deviation $\sigma = 4.5$, then the value of the t -test statistic is:

1. 2.14
2. - 1.54
3. 0.84
4. 1.988

Options 1. 1

2. 2
3. 3
4. 4

Q.50 If the corner points of the bounded feasible region for a Linear Programming Problem (LPP) are $A(0, 2)$, $B(3, 0)$, $C(2, 3)$ and $D(3, 1)$, then the maximum value of the objective function $Z = 4x + 2y$ occurs at

1. $(0, 2)$ only
2. the mid-point of the line segment joining the points $(2, 3)$ and $(3, 1)$ only
3. $(2, 3)$ and $(3, 1)$ only
4. every point on the line segment joining the points $(2, 3)$ and $(3, 1)$

Options 1. 1

2. 2
3. 3
4. 4

Section : Core Mathematics

Q.51 $\int \tan^{-1} \sqrt{x} dx$ equals to: (Here C is an arbitrary constant)

1. $(x + 1) \tan^{-1} \sqrt{x} - \sqrt{x} + C$
2. $x \tan^{-1} \sqrt{x} - \sqrt{x} + C$
3. $\sqrt{x} - x \tan^{-1} \sqrt{x} + C$
4. $\sqrt{x} - (x + 1) \tan^{-1} \sqrt{x} + C$

Options 1. 1

2. 2
3. 3
4. 4

Q.52 If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{j} - \hat{k}$ and $\vec{c} = \hat{i} + \hat{j} + \hat{k}$ such that $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$, then $\vec{r}$ is:

1. $\hat{i} + 3\hat{j} - \hat{k}$
2. $\hat{i} - 3\hat{j} + \hat{k}$
3. $3\hat{i} + \hat{j} - \hat{k}$

4. $\hat{i} - \hat{j} + 3\hat{k}$

Options 1. 1

2. 2

3. 3

4. 4

Q.53

If $y^{\frac{1}{m}} + y^{-\frac{1}{m}} = 2x$, then the value of $(x^2 - 1)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$ is:

1. 0

2. my

3. m^2y

4. m^2

Options 1. 1

2. 2

3. 3

4. 4

Q.54

If A and B are independent events, then which of the following statements are TRUE?

(A) $P(A \cap B) = P(A) \cdot P(B)$

(B) $P(A \cap B) = P(A) - P(B)$

(C) $P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$

(D) $P(A \cap B) = P(A) \cdot P(B|A)$

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (C) only

2. (B) and (C) only

3. (C) and (D) only

4. (A), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.55

Distance of the point $(2, 4, -1)$ from the line $\frac{10+2x}{2} = \frac{y+3}{4} = \frac{6-z}{9}$ is

1. $\sqrt{14}$ units

2. 13 units

3. 7 units

4. $\sqrt{7}$ units

Options 1. 1

2. 2

3. 3

4. 4

Q.56

If the line $\frac{-x+1}{3} = \frac{-y-2}{-2k} = \frac{z+3}{2}$ and $\frac{-1+x}{3k} = \frac{-1+y}{1} = \frac{-z+6}{5}$ are perpendicular, then the value of k is:

1. $\frac{10}{7}$

2. $-\frac{10}{7}$

3. $\frac{1}{2}$

4. $\frac{4}{3}$

Options

1. 1

2. 2

3. 3

4. 4

Q.57 Match **List-I** with **List-II**

List-I

List-II

(A) $\int_{-a}^a f(x)dx = 0$

(I) 0

(B) $\int_0^{2a} f(x)dx = 2 \int_0^a f(x)dx$

(II) 1

(C) $\int_{-\pi}^{\pi} \cos x dx$

(III) f is an odd function

(D) $\int_{-1}^1 x^{101} dx + 1$

(IV) $f(2a - x) = f(x)$

Choose the **correct** answer from the options given below:

1. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

2. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

3. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

4. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

Options

1. 1

2. 2

3. 3

4. 4

Q.58 A letter is known to have come either from KOLKATA or TATANAGAR. On the envelope just two consecutive letters TA are visible. The probability that letter has come from TATANAGAR is

1. $\frac{2}{5}$

2. $\frac{3}{5}$

3. $\frac{1}{4}$

4. $\frac{2}{3}$

Options 1. 1

2. 2

3. 3

4. 4

Q.59 If $\begin{vmatrix} 1 & -2 & 5 \\ 2 & a & -1 \\ 0 & 4 & 2a \end{vmatrix} = 86$, then product of all values of a is:

1. 21

2. - 21

3. 10

4. - 4

Options 1. 1

2. 2

3. 3

4. 4

Q.60 Bag A contains 2 unbiased and 3 biased coins whereas Bag B contains 3 unbiased and 2 biased coins. A bag is selected at random and 2 coins are taken out simultaneously. The probability, that both coins are unbiased is:

1. $\frac{1}{5}$

2. $\frac{1}{10}$

3. $\frac{2}{5}$

4. $\frac{3}{10}$

Options 1. 1

2. 2

3. 3

4. 4

Q.61 The minimum value of the function $f(x) = x^3 + (10 - x)^3$ occurs at:

1. $x = 0$

2. $x = 3$

3. $x = 5$

4. $x = \frac{10}{3}$

Options 1. 1

2. 2

3. 3

4. 4

Q.62

If $x = e^{\cos 2t}$, $y = e^{\sin 2t}$, then $\frac{dy}{dx}$ equals to

1. $\frac{y \log_e x}{x \log_e y}$

2. $\frac{x \log_e x}{y \log_e y}$

3. $-\frac{y \log_e x}{x \log_e y}$

4. $-\frac{x \log_e x}{y \log_e y}$

Options 1. 1

2. 2

3. 3

4. 4

Q.63

If $A = \begin{bmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{bmatrix}$, then the value of $\det(\text{adj}(2A))$ is:

1. 100

2. 320

3. 400

4. 1600

Options 1. 1

2. 2

3. 3

4. 4

Q.64

Match List-I with List-II

List-I

List-II

Mathematical Statement Value

(A) $\hat{i} \cdot (\hat{j} \times \hat{k})$

(I) $-\hat{k}$

(B) $\hat{j} \cdot (\hat{i} \times \hat{k})$

(II) 1

(C) $\hat{i} \times (\hat{j} \times \hat{k})$

(III) -1

(D) $\hat{j} \times \hat{i}$

(IV) $\vec{0}$

Choose the \textbf{correct} answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

2. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

3. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

4. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

Options

1. 1

2. 2

3. 3

4. 4

Q.65

If $2f(x) + f\left(\frac{1}{x}\right) = x^2 + 1$, then $\int f(x)dx$ is: (Here C is an arbitrary constant)

1. $\frac{1}{3} \left(\frac{2}{3}x^3 + \frac{1}{x} + x \right) + C$

2. $\frac{2}{3}x^3 - \frac{1}{x} + x + C$

3. $\frac{x^3}{3} + x + C$

4. $\frac{1}{3} \left(\frac{2}{3}x^3 - x \right) + C$

Options

1. 1

2. 2

3. 3

4. 4

Q.66

Match \textbf{List-I} with \textbf{List-II}

List-I

List-II

Differential Equation Integrating Factor

(A) $ydx + (x - y^3)dy = 0$ (I) e^{-x}

(B) $x \frac{dy}{dx} + y = x^2$ (II) $\frac{1}{x}$

(C) $\frac{dy}{dx} - y = e^x$ (III) y

(D) $xdy - ydx = x^3dx$ (IV) x

Choose the \textbf{correct} answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

2. (A) - (IV), (B) - (II), (C) - (I), (D) - (III)

3. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

4. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

Options 1. 1

2. 2

3. 3

4. 4

Q.67 The area (in square units) of the region bounded by the curves $3y^2 = ax$, $y = a$, $a > 0$ and y-axis is:

1. a

2. $3a$

3. $2a^2$

4. a^2

Options 1. 1

2. 2

3. 3

4. 4

Q.68 Match \textbf{List-I} with \textbf{List-II}

List-I

List-II

(A) $\cos^{-1} x + \cos^{-1}(-x)$ (I) $\frac{\pi}{3}$

(B) $\operatorname{cosec}^{-1}(-x) + \sec^{-1}(-x)$ (II) $-\frac{\pi}{3}$

(C) $\tan^{-1} \sqrt{3} - \sec^{-1}(-2)$ (III) π

(D) $\tan^{-1} \left(\tan \frac{4\pi}{3} \right)$ (IV) $\frac{\pi}{2}$

Choose the \textbf{correct} answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

2. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

3. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

4. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

Options 1. 1

2. 2

3. 3

4. 4

Q.69 The relation R in $\mathbb{R}$ (set of real numbers) is defined by $R = \{(a, b) : a \leq b^3\}$, then R is

1. Reflexive relation

2. Transitive but not symmetric

3. Reflexive and transitive but not symmetric

4. Neither reflexive nor symmetric nor transitive

Options 1. 1

2. 2

3. 3

4. 4

Q.70

A function $f: R \rightarrow \{x \in R: -1 < x < 1\}$ is defined as $f(x) = \frac{x}{1+|x|}$, then f is:

1. neither one-one nor onto
2. one-one only
3. onto only
4. both one-one and onto

Options 1. 1

2. 2

3. 3

4. 4

Q.71

The shortest distance between lines $\frac{-x-3}{4} = \frac{y-6}{3} = \frac{z}{2}$ and $\frac{-x-2}{4} = \frac{y}{1} = \frac{z-7}{1}$ is:

1. $\frac{1}{9}$ units

2. 81 units

3. 9 units

4. 7 units

Options 1. 1

2. 2

3. 3

4. 4

Q.72

If $|\vec{a}| = 1$, $|\vec{b}| = 2$, $|2\vec{a} + \vec{b}| = 2\sqrt{3}$ then $|\vec{a} - \vec{b}|$ is:

1. $\sqrt{3}$

2. $\sqrt{5}$

3. $\frac{2}{\sqrt{3}}$

4. $\frac{2}{\sqrt{5}}$

Options 1. 1

2. 2

3. 3

4. 4

Q.73

If C_{ij} represents the cofactor of element a_{ij} of the matrix $A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 2 & 0 \\ 4 & 1 & 5 \end{bmatrix}$, then the value of $C_{23} + C_{31} - C_{22}$ is

1. 14
2. -4
3. -10
4. 8

Options

1. 1
2. 2
3. 3
4. 4

Q.74

If matrix $A = \begin{bmatrix} p & -3 \\ -4 & p \end{bmatrix}$ and $|A^3| = 64$, then the value of p is:

1. ± 1
2. ± 2
3. ± 3
4. ± 4

Options

1. 1
2. 2
3. 3
4. 4

Q.75

The value of k for which the function $f(x) = \begin{cases} \frac{1-\cos 8x}{16x^2} & , \text{ if } x \neq 0 \\ k & , \text{ if } x = 0 \end{cases}$ is continuous at $x = 0$ is:

1. 0
2. 2
3. -2
4. 1

Options

1. 1
2. 2
3. 3
4. 4

Q.76

Area of region bounded by the curves $x = y^3$, $x = 0$ between $y = -1$ and $y = 2$ is:

1. $\frac{15}{4}$ sq. units
2. $\frac{17}{4}$ sq. units
3. $\frac{19}{4}$ sq. units

4. $\frac{21}{4}$ sq. units

Options 1. 1

2. 2

3. 3

4. 4

Q.77

If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then the value of $A^2 - 5A + 6I$ is

1. $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

2. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

3. $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

4. $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

Options 1. 1

2. 2

3. 3

4. 4

Q.78

The semi vertical angle of a right circular cone of maximum volume of a given slant height is

1. $\tan^{-1} 2$

2. $\sin^{-1} \left(\frac{1}{3} \right)$

3. $\cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$

4. $\sin^{-1} \left(\frac{1}{\sqrt{3}} \right)$

Options 1. 1

2. 2

3. 3

4. 4

Q.79

The corner points of the bounded feasible region determined by the system of linear inequalities are $(0, 0)$, $(2, 4)$, $(0, 5)$

and $(4, 0)$. If the maximum value of $Z = ax + by$, where $a, b > 0$ occurs at both $(2, 4)$ and $(4, 0)$, then

1. $3a = b$

2. $a = b$

3. $a = 2b$

4. $2a = b$

Options 1. 1

2. 2

3. 3

Q.80 Match \textbf{List-I} with \textbf{List-II}

Let A and B be any two events

List-I **List-II**

(A) $P(A')$ (I) $\frac{P(A \cap B)}{P(A)}; P(A) \neq 0$

(B) $P(\Phi)$ (II) $\frac{P(A \cap B)}{P(B)}; P(B) \neq 0$

(C) $P(A|B)$ (III) $1 - P(A)$

(D) $P(B|A)$ (IV) 0

Choose the \textbf{correct} answer from the options given below:

1. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

2. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

3. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

4. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.81 The function $f(x) = \sin 3x, x \in \left[0, \frac{\pi}{2}\right]$

(A) is increasing on $\left[0, \frac{\pi}{6}\right]$

(B) is decreasing on $\left[\frac{\pi}{6}, \frac{\pi}{2}\right]$

(C) is increasing on $\left[0, \frac{\pi}{2}\right]$

(D) is decreasing on $\left[0, \frac{\pi}{2}\right]$

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (C) only

2. (A) and (B) only

3. (B), (C) and (D) only

4. (D) and (A) only

Options 1. 1

2. 2

3. 3

4. 4

Q.82 Solution of the differential equation

$$\frac{dy}{dx} = \sqrt{1+x^2+y^2+x^2y^2} \text{ is : (Here C is an arbitrary constant)}$$

1. $\log \left| \frac{y + \sqrt{1+y^2}}{x + \sqrt{1+x^2}} \right| = C$

2. $\log \left| \frac{y + \sqrt{1+y^2}}{\sqrt{x + \sqrt{1+x^2}}} \right| = C$

3. $\log \left| \frac{y + \sqrt{1+y^2}}{\sqrt{x + \sqrt{1+x^2}}} \right| = \frac{x}{2} \sqrt{1+x^2} + C$

4. $\log \left| \frac{y + \sqrt{1+y^2}}{x + \sqrt{1+x^2}} \right| = \frac{x}{2} \sqrt{1+x^2} + C$

Options 1. 1

2. 2

3. 3

4. 4

Q.83 If a matrix P is both symmetric and skew-symmetric, then

1. P is a diagonal matrix

2. P is a square matrix

3. P is a zero matrix

4. P is an identity matrix

Options 1. 1

2. 2

3. 3

4. 4

Q.84 In a linear programming problem, the constraints on decision variables x and y are $y - 2x \leq 0, y \geq 0, 0 \leq x \leq 5$. The feasible region of the above problem:

1. is bounded in the first quadrant

2. is unbounded in the first quadrant

3. is unbounded in first and second quadrants

4. does not exist

Options 1. 1

2. 2

3. 3

4. 4

Q.85 If $\vec{a} = 3\hat{i} - 6\hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} - 4\hat{j} + \lambda\hat{k}$ are such that $\vec{a} \parallel \vec{b}$, then $3\lambda + 2 =$

1. 0

2. 2

3. 4

4. 6

Options 1. 1

2. 2

3. 3

4. 4

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